

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	i		Dependent Variable = numbers of each species (1) Independent Variable = (presence of) pollution (1)	2			2		2
		ii		Some easier to catch than others / misidentification/ recounting/ they move around/ camouflaged	1			1		1
		iii		Calculation of $N(N-1) = 22350$ (1) Calculation $\sum n(n-1) = 6926$ (1) calculation diversity index = 0.69 (1)		3		3	3	
		iv		biodiversity was low(er)/ the {number/types} of species were low(er).		1		1		
		v		There had been a mathematical error (1) highest possible diversity index is 1 (1)		2		2	2	
	(b)	i		Any two (x1) from: The existence of a number of distinct (inherited) varieties (coexisting in the same population in a single species)/ snails {are different colours/ have different bands}/ different morphology/ different phenotypes(1) at frequencies too great to be explained by recurrent mutation (1) multiple alleles for the same gene(1)	2			2		

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3	b	ii		• Different {colours / banding/ features} give (a selective) advantage in different habitats/ different colours are camouflaged in different habitats (1) • Habitats vary depending on times of the {year / seasons} and so different colours will have an advantage (1) • {Main predator / thrush} will predate different forms of the snail in {different areas / different seasons} (1)		3		3		
				Question 3 total	5	9	0	14	5	3